PROJECT

7.6

INFRASTRUCTURE AS CODE WITH CLOUDFORMATION

DEVOPS SERIES

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Objective

This project uses AWS CloudFormation to automate the creation and management of infrastructure as code. It ensures consistent, reusable, and version-controlled resource provisioning.

It sets up a CI/CD pipeline using CodeArtifact, CodeBuild, and CodeDeploy, enabling smooth deployment from development to production all managed through CloudFormation templates.

Services

Amazon EC2 - Hosts the deployed web application.

AWS CodeArtifact - Stores and manages application dependencies.

AWS CodeBuild - Compiles source code and runs tests in a repeatable build environment.

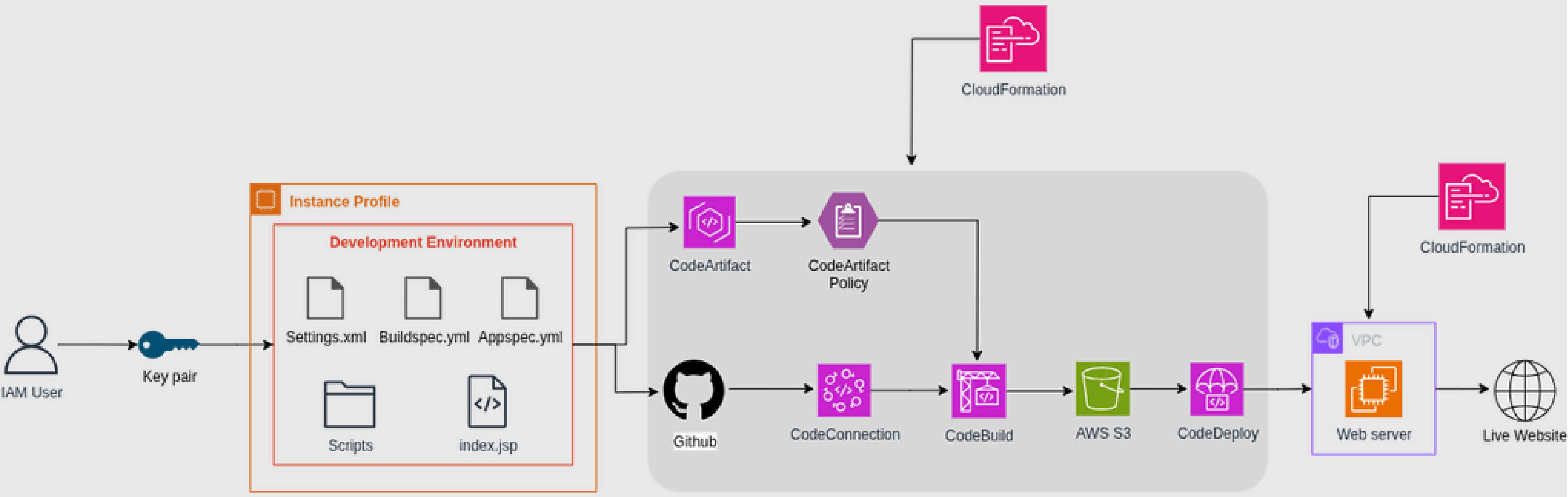
AWS CodeDeploy - Automates deployment to Amazon EC2 instances.

AWS CloudFormation - Automates provisioning of infrastructure using templates.

AWS IAM - Manages roles and permissions for CodeBuild, CodeDeploy, and CloudFormation.

AWS Secrets Manager - Secures and manages secrets added to the template

Architecture Diagram



This architecture uses AWS CloudFormation to automate the deployment pipeline. Code is pulled from GitHub, CodeArtifact handles dependencies, and CodeBuild compiles the app. Artifacts are stored in S3 and deployed to a web server in a VPC via CodeDeploy. All components are provisioned through CloudFormation for consistency and automation.

Steps to Implement

1. Set up your development EC2 instance:
Launch an EC2 instance with necessary tags, security groups, and IAM roles to serve as your application host and enable future CodeDeploy targeting.

2. Create a CodeArtifact repository:
Set up a CodeArtifact domain and repository to store your application's dependencies securely. Configure access permissions using IAM policies.

3. Set up your CodeBuild project:
Create a CodeBuild project that pulls code from your source repository, installs dependencies from CodeArtifact, and builds the application in a clean environment.

4. Set up your CodeDeploy application and group:
Create a CodeDeploy application, define a deployment group, attach your EC2 instance using tags, and configure rollback settings for safe updates.

5. Generate a CloudFormation template:
Write a CloudFormation template that defines all resources, EC2, IAM roles, CodeArtifact, CodeBuild, and CodeDeploy, to automate setup and ensure consistency.

6. Test your CloudFormation stack:
Deploy your CloudFormation stack to validate the configuration. Verify that resources are correctly provisioned and the CI/CD pipeline functions as expected.

7. Add secret resources using AWS Secrets Manager:
Securely manage environment variables, credentials, or tokens by referencing Secrets Manager in your CloudFormation template to keep sensitive data safe.

Story Time

CloudFormation (The Magical Blueprint Scroll):
CloudFormation is your enchanted scroll that brings your entire kingdom to life. Just write your wishes (in YAML/JSON), and poof!, your infrastructure appears like magic, always the same, always perfect.

EC2 (The Cloud Castle for Your App):
EC2 is your grand castle in the cloud where your app is crafted. You launch it, equip it with tools, and access it from afar like a wizard using VSCode's Remote-SSH portal.

CodeArtifact (The Potion Library):
CodeArtifact is your mystical library of potions (dependencies). You fetch all the magical ingredients your app needs, safely stored and versioned for future spellcasting.

CodeBuild (The Spell Caster):
CodeBuild is the spell caster that takes your raw code and potion ingredients and brews it into something powerful. It compiles, tests, and builds your code with precision.

Amazon S3 (The Sacred Archive):
S3 is your sacred vault where all the magical artifacts (build outputs) are stored. These relics are later used to deploy your application across the realm.

CodeDeploy (The Sky Messenger):
CodeDeploy is your flying messenger that takes the built artifact from the vault and delivers it straight to your cloud castle (EC2). It handles rollbacks too, just in case your magic misfires.

BY
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THANK YOU!

DEVOPS ENGINEER
IN THE MAKING

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